
The Architecture of Errors: From Universal Impossibility to Patch-Local LLM Reliability

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Abstract

Universal LLM reliability is not a finite-library problem: across all possible tasks, tools, schemas, knowledge sources, and evaluator expectations, new intervention-distinguishable failure modes can appear without bound, so no finite intervention dictionary can guarantee bounded residual error universally. But deployed systems do not operate over the whole universe; they operate inside bounded patches (legal review, medical RAG, code repair, customer-support agents, contract extraction) with recurring tasks, schemas, tools, and evaluator expectations. Within such patches, empirical evidence suggests failures are sparse, repetitive, and concentrated in a small recurring catalogue, so reliability becomes a local catalogue-discovery and intervention-coverage problem rather than an exponential token-length problem. We make this transition formal with two propositions: a negative result that no universal finite intervention dictionary exists, and a positive result that under logarithmic or capped mode discovery the patch-local intervention budget grows polylogarithmically in sequence length and becomes domain-constant once the patch catalogue saturates. The framework relocates rather than dissolves long-context difficulty: where the number of hard decisions itself grows with task length, reliability remains hard; the contribution is to identify the on-axis intervention rather than to make those regimes easy.

1 Introduction

The standard worry about long-context autoregressive generation is exponential. If every token has independent error probability e , then after n tokens the chance of a fully correct output is $(1 - e)^n$, which collapses to zero for any nontrivial e and large enough n [LeCun, 2023, Dziri et al., 2023]. Earlier work in this series [Anonymous, 2025] argued that this worry is misplaced because errors are not distributed uniformly across tokens: only 5–10% of tokens are “key” (genuinely dependent on long-range context [Fang et al., 2025]), while the remaining majority become nearly deterministic once enough context accumulates. Replacing $(1 - e)^n$ with the two-rate model $P(\text{correct}) = (1 - e_{\text{key}})^k (1 - e_{\text{non}})^{n-k}$ with $k \ll n$ scaling sublinearly in n recovers the observed long-context coherence and converts the reliability question from “how does n grow?” to “how does k grow?”.

From sparse tokens to recurring patterns. This paper takes the next step. Sparsity tells us *where* errors live; the natural follow-up is *what* they are. Recent failure-mode atlases supply a striking empirical answer: errors are not only sparse, they are also *repetitive*. ErrorAtlas [Ashury-Tahan et al., 2026], covering 83 models across 35 datasets, organises observed failures into 17 named categories sorted by prevalence, with a long-tailed distribution heavily concentrated in the head. In code, two

error types (AssertionError and NameError) cover 86.35% of failures on HumanEval across 14 LLMs, replicated across 23 models on HumanEval Pro and MBPP Pro [Wen et al., 2024]. In math, MWPEs-300K [Sun et al., 2025] categorises 304,865 errors from 15 LLMs across four math word-problem datasets and reports that dataset characteristics shape error patterns systematically. The same picture appears in multi-hop QA [Zhang et al., 2026], agentic tool use [Cemri et al., 2025], and retrieval-augmented generation [Wood and Forbes, 2024]. These observations suggest a third layer in the reliability architecture, complementing key-token sparsity (Layer 1) and within-key stratified-manifold structure (Layer 2 in Anonymous 2025). Within the small set of key tokens, only a fraction β produce *hard* failures, and inside bounded deployment patches those hard failures appear to cluster into a finite or effectively capped catalogue of recurring modes whose size grows much more slowly than the number of observed events.

Contributions. The central result is the conditional polylog intervention budget of Proposition 2: $m \geq \lceil |C|^{1-\varepsilon/e_{\text{hard}}} \rceil$, with a doubly-logarithmic rate as the optimistic special case under the postulate above. The *sequence*-level analogue is strictly tighter and approaches full-catalogue coverage as k grows; we surface this asymmetry rather than bury it. Around this central claim the paper does four supporting things. The three-layer formal framework, comprising sparsity (α), stratification (β), and a bounded mode catalogue ($|C|$), partitions key tokens into easy and hard subsets, then adds the recurring-pattern catalogue (§3). The empirical postulate of logarithmic mode discovery is calibrated against ErrorAtlas, HumanEval, and MWPEs, with $\sigma \in [0.87, 1.85]$ across anchors and $\sigma \approx 1.85$ adopted from ErrorAtlas as the conservative target. Section 4 synthesises evidence for clustering, cluster-selective interventions, and sublinear length scaling, drawing on ~ 60 prior published results including the six-axis capability-elimination harvest of 28 quantitatively-anchored citations stratified into Patterns A/B/C (Appendix F). And the re-audit of the most-cited steep-decay results (Appendix G) shows each decays over a variable other than raw token length.

2 Related Work

Failure-mode taxonomies. ErrorAtlas [Ashury-Tahan et al., 2026], MWPEs-300K [Sun et al., 2025], the HumanEval categorisation of Wen et al. [2024], and the RFMDataset [Guo et al., 2025] together suggest that, at any given corpus size, the number of distinct named failure modes is small (typically 8–20) and the cumulative coverage of the top modes is high. Domain-specific taxonomies for multi-hop QA [Zhang et al., 2026], multi-agent systems [Cemri et al., 2025], and tool-augmented agents [Yao et al., 2024] report the same pattern.

Targeted interventions. Each named cluster has been the subject of focused intervention research. Python execution closes most of the arithmetic-error cluster [Gao et al., 2023a, Chen et al., 2023, Gou et al., 2024b]; constrained decoding eliminates format violations [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]; execution feedback closes most code-logic errors [Shinn et al., 2023, Huang et al., 2023]; process reward models reduce step-level reasoning errors [Wang et al., 2024b, Lightman et al., 2024]; RAG strongly reduces hallucinations [Wood and Forbes, 2024]; preference optimisation closes over-refusal [Karaman et al., 2024]; structured uncertainty fixes most tool-call failures [Suri et al., 2025]. Two structural observations recur: each intervention is *cluster-selective* (residuals belong to a different named cluster, not to the targeted one), and *additivity is approximate but not perfect* [Patel et al., 2026, Le, 2026].

Length-decay benchmarks. A parallel literature measures the *shape* of the reliability decay curve. Mild-decay results (Loong [Wang et al., 2024a], GSM- ∞ [Zhou et al., 2025], RULER [Hsieh et al., 2024], anchor-based LLMs [Pang et al., 2024]) report log-linear, sigmoidal, or threshold-like decay inconsistent with smooth $(1 - \varepsilon)^n$. Steep-decay results [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026] are sometimes cited as exponential-compounding evidence. The apparent tension is reduced on careful reading: every steep-decay paper decays over a variable distinct from raw token length, which Appendix G documents.

Gap. What is missing is a quantitative bridge between the small, repeating taxonomy catalogue and a polylog intervention budget. We provide that bridge in §3.

85 3 Theoretical Framework

86 **Four levels of $|C|$.** Before any proposition, a definitional setup. LLM error analysis routinely
87 conflates four objects that the rest of this paper needs to keep apart:

- 88 • **L1: failure events.** Concrete LLM errors observed in a benchmark; the raw data behind any
89 taxonomy.
- 90 • **L2: empirical taxonomy categories.** Researcher-chosen labels grouping L1 events. ErrorAt-
91 las’s 17 categories, MWPE’s top-4 classes, and HumanEval’s `AssertionError/NameError`
92 partition all live here. L2 is observable but depends on the taxonomer’s resolution choices.
- 93 • **L3: latent failure modes.** The unobserved underlying clusters that L2 approximates. L3 is the
94 quantity Postulate 1 is morally about, but it cannot be measured directly; we treat L2 as a noisy
95 proxy for L3.
- 96 • **L4: capability axes / interventions.** The engineering unit: a Python interpreter, a constrained
97 decoder, a retrieval-augmented generator. L4 is coarser than L2 in practice; one capability axis
98 typically targets several L2 categories at once (§4, Claim B).

99 Throughout this section, $|C|$ refers to the **L2** count: that is what published taxonomies report.
100 Postulate 1 is therefore *engineering*-relevant because L4 is coarser than L2 (so a small intervention
101 library can sweep across many categories at once), and *epistemically* conditional because L2 is a noisy
102 proxy for the actual L3 mode catalogue whose faithfulness has not been independently measured.
103 Tagging the formal claims at the right level keeps the framework honest where casual error-talk slips.

104 **Roadmap.** The framework separates three questions that are often conflated. First, *where* do errors
105 occur? Only a sparse subset of key decisions. Second, *what* kinds of errors recur? A local catalogue
106 of failure modes. Third, *what fixes them*? A smaller library of capability interventions. The two
107 propositions of §3.4 formalise the transition from universal impossibility to patch-local tractability:
108 there is no finite intervention dictionary that covers all possible deployments, but inside a fixed
109 deployment patch the required library is small and slowly growing.

110 3.1 β -stratification of key tokens

111 The two-rate model of Anonymous [2025] distinguishes k key tokens (error rate e_{key}) from $n - k$
112 non-key tokens ($e_{\text{non}} \ll e_{\text{key}}$). Empirical atlases suggest that even within the key-token class errors
113 are not uniformly distributed: most key tokens are “decisions” for which the model has stable
114 representations; only a fraction concentrate the actual failures.

115 **Definition 1** (Hard fraction). Partition the k key tokens of a sequence into easy and hard subsets,

$$k_{\text{hard}} = \beta k, \quad k_{\text{easy}} = (1 - \beta)k, \quad \beta \in (0, 1),$$

116 where hard key tokens have an elevated error rate e_{hard} corresponding to manifold-transition decisions
117 in the sense of Anonymous [2025, §3.2], and easy key tokens have $e_{\text{easy}} \approx e_{\text{non}}$.

118 Under Definition 1, the composed sequence-level reliability becomes

$$P(\text{correct}) = (1 - e_{\text{hard}})^{\beta k} (1 - e_{\text{easy}})^{(1-\beta)k} (1 - e_{\text{non}})^{n-k}. \quad (1)$$

119 We do not assume iid token errors: the three rates summarise *conditional per-decision failure*
120 *hazards* after conditioning on prior decisions being correct, and grouping by stratum yields the
121 multiplicative survival expression. Because $e_{\text{easy}} \approx e_{\text{non}}$ once context accumulates, the failure rate
122 is dominated by the βk hard tokens, and we treat e_{hard} as the load-bearing quantity. The parameter
123 β is *latent*: failure-mode atlases report $\Pr(\text{category} \mid \text{error occurred})$, which does not identify
124 $\Pr(\text{hard} \mid \text{key-token decision})$. We carry β symbolically.

125 3.2 Empirical postulate: logarithmic mode discovery

126 Stratification names a target, the βk hard-token decisions, but does not yet tell us how many distinct
127 ways those hard decisions can fail. The empirical answer from Section 1 is that failures repeat: a
128 small number of recurring patterns covers most of the mass. The next question is how the catalogue’s

size grows with the number of observed failures, because that is the quantity an intervention library has to keep up with.

Two candidates exist in the type-token literature: Heaps’ law (power-law, $|C| \approx K \cdot k_{\text{hard}}^b$ with $b \in [0.4, 0.6]$ [Manning et al., 2008]), and logarithmic discovery. **Zipfian rank-frequency does not imply logarithmic growth.** Heaps’ law is the correct type-token consequence of Zipf-distributed events, and it is power-law, not logarithmic. We therefore state logarithmic mode discovery as an empirical postulate, defensible by direct measurement of error taxonomies, not as a theorem.

Two sample-size variables matter, and they are easy to conflate. Let $h(n) = \beta k(n)$ denote the number of hard-token decisions in a single sequence of length n ; let T denote the number of observed hard-failure events in a corpus used to discover the domain catalogue. These are not the same object: $h(n)$ controls per-sequence exposure, T controls empirical discovery. We write C_D for the full reachable catalogue inside domain patch D , $C_{\text{seen},D}(T)$ for the subset discovered after T sampled failures, and $C_{\text{active},D}(n)$ for the subset a single sequence of length n can activate.

Postulate 1 (Patch-indexed catalogue discovery). Within a fixed application domain D , the number of named recurring failure modes discovered after T sampled hard-failure events is bounded by

$$|C_{\text{seen},D}(T)| \leq \min(A_D + \sigma_D \ln T, |C_D|), \quad \sigma_D > 0, A_D \geq 0.$$

The cap $|C_D|$ is the domain-imposed ceiling. The postulate concerns catalogue *discovery*, not the number of hard decisions inside a single sequence.

Assumption 2 (Per-sequence active-mode exposure). For a single sequence of length n in domain D ,

$$|C_{\text{active},D}(n)| \leq \min(A'_D + \sigma'_D \ln h(n), |C_D|).$$

Primed constants are distinct from those of Postulate 1: corpus discovery and per-sequence activation need not share the same rate.

The two bounds answer different questions, and the rest of the paper picks the right one at each point. Proposition 2’s sequence-length scaling rate is governed by $C_{\text{active},D}(n)$ via Assumption 2: how much of the catalogue can a single sequence touch? Full domain-library budgeting, the practical engineering target for a deployed system, uses C_D : how large does the library need to be to cover the domain’s reachable failure modes? The cap $|C_D|$ enters both: once enough has been sampled (or activated) to saturate the domain ceiling, further growth stops and the budget becomes a domain-constant function of $|C_D|$ alone.

Empirical defense. Three published taxonomies anchor the mode-rate parameter σ at endpoint corpus scales. ErrorAtlas at $T \gtrsim 10^4$ observed failures has $|C| = 17$, giving $\sigma \approx 17 / \ln T \approx 1.85$ at $T = 10^4$ under $A = 0$. HumanEval [Wen et al., 2024], on a comparable corpus size, has $|C| \in [8, 12]$ and accordingly $\sigma \in [0.87, 1.30]$. MWPEs, an order of magnitude larger ($T \approx 3 \times 10^5$), still has $|C| \in [15, 20]$ and yields $\sigma \in [1.2, 1.6]$. Across the three anchors, σ falls in the range $[0.87, 1.85]$; we carry $\sigma \approx 1.85$ (the ErrorAtlas value, the largest of the three) as a conservative downstream target. Endpoint counts at a single corpus scale are not discovery curves: a subsample-vs-discovered-modes measurement has not been published for any LLM error taxonomy at this writing, and remains the explicit empirical test we flag (§6, Appendix E).

Capabilities are coarser than error classes. Postulate 1’s $\sigma \approx 1.85$ is calibrated against *named error categories*, but a single capability axis often targets several L2 categories at once. A Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; a constrained decoder eliminates by construction the format-violation cluster and the structural component of “missing required element.” Postulate 1 is therefore a conservative input for the engineering question; the formal coverage problem is weighted set cover, with the ranked-category form of Proposition 2 as a tractable proxy. An alternative power-law (Heaps) formulation and symbolic-form sensitivity analysis across candidate cluster-count laws are given in Appendix E.

Domain patches cap the catalogue. A model deployed in a fixed application domain occupies a bounded sub-region of the stratified manifold, a *domain patch* P_D [Anonymous, 2025, Li and Sarwate, 2025]. Empirical support: embedding-space intrinsic dimensions of ~ 5 –15 across domains, well below ambient $\sim 10^3$ [Park et al., 2024, Li and Sarwate, 2025]; cross-domain accuracy spreads of 20–70% on a single base model [Wang et al., 2024c]; log-popularity thresholds below which

scaling fails [Mallen et al., 2023, Kandpal et al., 2023]. None of these *measure* $|C_D|$ directly; together they motivate the patch-indexed hypothesis. The patch makes logarithmic or saturating discovery dynamics plausible inside fixed domains; σ_D , A_D , β_D , $|C_D|$ are all domain-indexed. We treat the patch’s reachable catalogue as finite or effectively capped: a modelling choice motivated by the heterogeneity evidence above, not a derivation from boundedness alone.

3.3 Coverage by a targeted intervention library

With the catalogue defined, the next question is how much residual error a library of m targeted interventions actually closes off. Given a catalogue of $|C|$ recurring modes, let $p_1 \geq p_2 \geq \dots \geq p_{|C|}$ denote the ranked empirical hard-error mass of those modes with $\sum_i p_i = 1$, so a library covering the top m modes removes cumulative error mass $\sum_{i=1}^m p_i$. We adopt the log-coverage form

$$F(m; |C|) = \min\left(1, \frac{\ln m}{\ln |C|}\right) \quad (2)$$

as a *postulated* parametric approximation to that cumulative top- m mass, not as a raw count of categories covered. Reading the formula: the first few interventions cover disproportionate mass (the head of the distribution is dense), then incremental coverage falls off as the library saturates the catalogue. For our ErrorAtlas anchor ($|C| = 17$, $m = 5$), $F(5; 17) \approx 56.8\%$; doubling the library to $m = 10$ pushes it to $F(10; 17) \approx 81.3\%$.

This is an empirical best-fit choice, not derived from Zipf. ErrorAtlas sorts its 17 categories by prevalence but does not publish a single cross-domain cumulative top- m curve at this writing; alternative cumulatives (Zipf–Mandelbrot, truncated power law, saturating exponential) match the same qualitative shape with comparable freedom. Appendix E shows the polylog conclusion of §3.4 survives across the family.

Two caveats matter for what comes next. (a) F tracks cumulative error mass across ranked L2 categories, not L4 capability axes deployed; in real deployments, where one capability often sweeps several L2 categories, residual error at a given m is generally lower than (2) predicts. (b) The form is intended for $m \geq 2$, $|C| \geq 2$; we adopt $F(0; |C|) := 0$ and $F(1; |C|) := p_1$ (the mass of the single most prevalent mode) since the log expression is not anchored at $m = 1$. After deploying the top- m library, the residual per-hard-token error rate is

$$e_{\text{res}}(m) = (1 - F(m; |C|)) e_{\text{hard}}. \quad (3)$$

3.4 From universal impossibility to patch-local reliability

The finite-catalogue claim is patch-local. It does not say that all possible LLM failures can be covered by a universal list of fixes. Across all possible tasks, tools, schemas, knowledge sources, workflows, and evaluator expectations, new intervention-distinguishable failures can keep appearing. In that setting, a finite intervention dictionary is not a well-posed reliability target.

The positive result begins only after a deployment patch D has been fixed. Inside a patch, the task family, schemas, tools, evaluator expectations, and admissible workflows recur. The engineering question then changes from “Can one library cover all possible LLM failures?” to “How large must the local library be to cover enough of the reachable failure catalogue C_D ?”

Proposition 1 (No Universal Finite Intervention Dictionary). *Fix a residual-error tolerance ε . If a domain D contains an infinite sequence of failures where each new failure is not covered by any finite intervention dictionary covering all earlier failures, then the ε -resolution failure catalogue C_D^ε is infinite. Consequently, no finite intervention dictionary can guarantee residual error below ε over all of D .*

Proof sketch. Each new failure is intervention-distinguishable from the failures before it. The sequence therefore generates infinitely many distinct intervention modes, and any finite dictionary must miss some later mode. The full proof, including the precise definition of mode-level coverage, is in Appendix A.

Engineering meaning. Universal reliability is not a finite-library problem. A fixed list of interventions cannot cover open-ended LLM use in general. The rest of the paper is therefore about

225 *patch-local* reliability, not universal reliability. Proposition 1 is the inoculation against a misread: we
 226 are not claiming a fixed list of ≈ 50 named patterns covers LLM use in general.

227 The positive result composes β -stratification (Definition 1), the active-mode bound (Assumption 2),
 228 and the log-coverage form (Eq. 2) into a local budget statement. The result is conditional engineering
 229 math, not an unconditional theorem about LLMs: it depends on the log-coverage approximation, the
 230 active-mode exposure assumption, and the patch hypothesis that C_D is finite or effectively capped.
 231 Appendix B states these conditions explicitly before the derivation.

232 **Proposition 2** (Patch-Local Intervention Budget). *Fix a deployment patch D . Let $\varepsilon \in (0, e_{\text{hard}})$ be the*
 233 *target per-hard-token residual error rate. Under the log-coverage approximation of §3.3, a library*
 234 *covering the dominant local modes satisfies the target once*

$$m \geq \left\lceil |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}} \right\rceil, \quad (4)$$

235 where C_{eff} is either the active catalogue $C_{\text{active},D}(n)$ touched by a single sequence, or the full
 236 reachable catalogue C_D of the deployment patch. If $C_{\text{active},D}(n)$ grows logarithmically with the
 237 number of hard decisions and $k(n) = \Theta(\log n)$, then the pre-cap intervention budget grows doubly-
 238 logarithmically in sequence length. Once the patch catalogue saturates, the required budget becomes
 239 domain-constant in $|C_D|$.

240 **Proof sketch.** The library removes cumulative hard-error mass $F(m; |C|)$. The residual hard-token
 241 error rate is therefore $(1 - F) e_{\text{hard}}$. Requiring this residual to be at most ε rearranges, after substituting
 242 the log-coverage form, to the stated bound. The full step-by-step derivation, including the pre-cap
 243 and cap regimes, is in Appendix B.

244 **Engineering meaning.** Once the patch is fixed, the problem changes. The relevant engineering
 245 question is no longer whether arbitrary future failures exist, but how quickly the reachable local
 246 catalogue is discovered and how much hard-error mass is removed by the top interventions. The
 247 intervention budget for per-hard-token reliability is small and slowly growing, domain-constant in
 248 the cap regime, with the exact size determined by the local discovery and rank-coverage curves rather
 249 than by a universal prior.

250 **Sequence-level caveat.** Proposition 2 bounds residual error per hard decision. A one-shot sequence-
 251 level target is strictly stricter because many hard decisions occur in one output: as k grows the
 252 allowable residual per hard decision shrinks, and the required library approaches full-catalogue
 253 coverage. In some regimes the non-hard-token error mass alone already exceeds the sequence-level
 254 budget, so hard-token interventions cannot meet the SLA by themselves. The full three-regime
 255 analysis (and the per-hard-token tolerance τ_{seq} that converts a sequence target into the Proposition 2
 256 bound) is in Appendix C.

257 A common engineering rule of thumb, “fix ≈ 50 patterns per domain $\rightarrow \approx 90\%$ error reduction,” is
 258 best understood as a per-hard-token planning prior. The analogous sequence-level claim requires
 259 correspondingly more interventions and tighter coverage of the tail.

260 We call these statements propositions rather than theorems because their force is conditional: they
 261 formalise the consequences of the paper’s modelling assumptions rather than deriving a universal
 262 law of LLM reliability. The doubly-logarithmic rate is the optimistic special case under logarithmic
 263 active-mode exposure; Appendix E gives the Heaps and saturation variants. Appendix D expands on
 264 the conditional reading.

265 4 Empirical Evidence

266 We collect evidence for three load-bearing claims: errors cluster into a small recurring set (A); each
 267 cluster is addressable by one targeted intervention (B); reliability decays sublinearly with output
 268 length (C).

269 **Claim A: Failure-mode clustering.** The strongest single result is ErrorAtlas [Ashury-Tahan
 270 et al., 2026]: 83 models $\times$ 35 datasets, $\approx 7,000$ failures, 17 named categories with a long-tailed,
 271 head-concentrated prevalence ordering. Domain-specific Pareto reproduces the shape inside each

domain: math (MWPEs top-4 categories dominate per model with strong cross-model overlap [Sun et al., 2025]), code (AssertionError+NameError = 86.35% on HumanEval, replicated across 23 models [Wen et al., 2024]), multi-hop QA (a single dominant “missing-evidence” mode [Zhang et al., 2026]), agentic tool use (under 20 recurring modes across two studies [Cemri et al., 2025, Yao et al., 2024]), and RAG (a handful of distinguishable hallucination types [Wood and Forbes, 2024]). The catalogue is also *stable* across models: ErrorAtlas reports a fixed category hierarchy across 83 models; RFMDataset [Guo et al., 2025] finds “strikingly similar failure mode distributions” across ten advanced reasoning models; EDIT [Dai et al., 2025] measures a $\approx 4.7\%$ key-step fraction stable across models. Schaeffer et al. [2025] provide the closest theoretical anchor, proving that the observed aggregate power-law scaling of LLM evaluations requires the per-problem success-rate distribution to be heavy-tailed near $p = 0$, structurally related to Postulate 1.

Claim B: Cluster-selective capability interventions. A dedicated harvest yields 28 capability-elimination citations across six independent axes (arithmetic, code execution, format/structure, perception/grounding, knowledge/RAG, verification), each axis independently confirmed by between three and nine citations. The full table and the three structural patterns (Pattern A by-construction, B strong-empirical-with-class-shift, C moderate-with-shift) appear in Appendix F.

Arithmetic is the cleanest case: PAL [Gao et al., 2023a] lifts GSM-Hard from 20.1% to 61.5%, and the residuals are problem-comprehension errors rather than arithmetic ones. Format violations admit something stronger still: constrained decoding zeroes invalid-token probability by construction [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]. Code-logic errors yield to execution feedback (Reflexion + AgentCoder push HumanEval pass@1 from 80% to 96.3%, with residuals in spec-misinterpretation [Shinn et al., 2023, Huang et al., 2023]); reasoning steps to process supervision (Math-Shepherd: Mistral-7B MATH 28.6% $\rightarrow$ 43.5% [Wang et al., 2024b]); hallucinations to RAG, where Acurai [Wood and Forbes, 2024] reports 100% elimination on RAGTruth, a benchmark-conditional empirical result rather than a general by-construction guarantee. Two further clusters complete the picture: POROver lifts the not-overrefused rate from 57.6% to 82.1% [Karaman et al., 2024], SAGE-Agent raises When2Call accuracy from 36.5% to 65.2% [Suri et al., 2025].

The class-shift signature, in which post-intervention residuals belong to structurally different classes, is the empirical content of the cluster-orthogonality claim that underwrites Proposition 2’s composition step. Capabilities are coarser than error classes: a single Python interpreter removes execution-error components from five named clusters; constrained decoding eliminates format and the structural component of “missing required element” jointly. A negative control sharpens the selectivity claim. DebugBench [Tian et al., 2024] reports that execution feedback works for syntax and reference errors and is explicitly “unhelpful for logic errors,” exactly where the framework predicts capability provisioning fails.

Claim C: Sublinear length scaling. Direct evidence for sublinear (rather than exponential) decay shows up across very different measurement designs. Loong [Wang et al., 2024a] reports GPT-4o on Chain-of-Reasoning declining 81.6% $\rightarrow$ 32.9% across the 10K $\rightarrow$ >200K context bins; log-linear over $\log L$ and not well explained by smooth independent per-token exponential decay. GSM- ∞ [Zhou et al., 2025] attacks the independent-error model from a different angle: exponentially more inference compute yields only *linear* AUC gains, and DeepSeek-R1 maintains 10%+ accuracy at 130 reasoning operations where $(1 - \epsilon)^{130}$ for any reasonable ϵ predicts near-zero. Press et al. [2023] find a $\approx 40\%$ compositionality gap that is approximately *constant* across the GPT-3 family, direct evidence against per-step independent compounding (which would predict the gap to grow with chain length).

Structural correlates of the same phenomenon appear elsewhere. Pang et al. [2024]’s Anchor LLMs achieve $\approx 99\%$ K/V reduction with only minor accuracy compromise: the effective number of distinct attention-key vectors a model needs is far smaller than n . Think-Prune-Train [Costello et al., 2025] raises Pass@1 while Pass@20 plateaus, the manifold-transition signature. RULER [Hsieh et al., 2024] finds threshold behaviour inconsistent with smooth $(1 - \epsilon)^n$. METR [Kwa et al., 2025] fits logistic-in-log-length, which is mathematically sublinear in length itself.

Self-consistency [Wang et al., 2023] is consistent with but does not *prove* clustered structure: Condorcet aggregation of iid Bernoulli chains can yield similar gain magnitudes, so we treat it as suggestive rather than diagnostic. Prominent steep-decay counter-evidence [Dziri et al., 2023, Ku-

326 ratov et al., 2024, Wan et al., 2026, Karpinska et al., 2024] decays over variables distinct from raw
 327 token length; per-paper re-audits showing the relocation pattern are in Appendix G.

328 5 Practical Implications

329 **Reliability engineering is local patch coverage.** Within a fixed domain patch, Proposition 2 makes
 330 reliability a small-catalogue engineering problem rather than an asymptotic scaling problem. A
 331 team should initially budget for a library on the order of tens of interventions, then refine from the
 332 local mode-discovery curve $C_{\text{seen},D}(T)$ and the empirical rank-frequency distribution. As a rule
 333 of thumb consistent with the head-mass figures in ErrorAtlas, HumanEval, and MWPEs, ≈ 50
 334 named interventions cover the bulk of the per-hard-token failure distribution in many measured
 335 domains. This is a planning prior calibrated to current taxonomies, not a universal constant; the local
 336 mode-discovery curve sets the actual library size for any given deployment. The same base model in
 337 cardiology RAG, legal contract drafting, and code review yields three different libraries because C_D ,
 338 A_D , σ_D all change with the deployment domain.

339 **Per-hard-token vs. sequence-level targets matter for SLA design.** Per-hard-token residual error,
 340 the right metric for continuous-correction systems, has a polylog budget. Sequence-level failure
 341 probability, the right metric for one-shot systems, is strictly stricter and approaches full-catalogue
 342 coverage as k grows. Production deployments should design SLAs to match the actual cost structure,
 343 not read the per-token result as the production SLA.

344 **Library design as engineering, not asymptotic theory.** An intervention library can be assembled
 345 module-by-module, in any order, as long as modules target distinct failure classes. Layer-separated
 346 interventions (constrained decoding + retrieval + process supervision + tool call) compose approxi-
 347 mately additively in our data; the Le [2026] caveat applies only to interventions sharing a prompt
 348 channel. Library construction is therefore highly parallelisable.

349 **Capability libraries are coarser than error-class libraries.** Five named math classes (arithmetic,
 350 units, counting, list manipulation, date arithmetic) collapse under one Python interpreter; three code
 351 classes (SyntaxError, NameError, most TypeError) collapse under one execution-feedback loop;
 352 format violations and the structural component of “missing required element” (together more than
 353 20% of ErrorAtlas) collapse under one constrained-decoder deployment. A team budgeting for
 354 ≈ 50 interventions per domain can expect to need fewer than 50 capability-axis interventions while
 355 covering ≈ 50 named clusters. The six axes of Appendix F are closer to the right unit of engineering
 356 accounting than the twelve clusters.

357 6 Discussion

358 **By-construction elimination as a boundary case.** A subset of the capability-elimination harvest
 359 deserves separate accounting. Seven of the 28 citations (Appendix F; constrained decoders [Suresh
 360 et al., 2025, Dong et al., 2025, Li et al., 2026, Zhang et al., 2023, OpenAI, 2024], proof kernels [Ren
 361 et al., 2025], static syntax checks) achieve residual error rate equal to zero *by mathematical construc-*
 362 *tion*, not statistically. Constrained decoders set $P(\text{invalid token}) = 0$ at every generation step; the
 363 class of grammar-violating outputs is mathematically empty. In this regime the polylog bound of
 364 Proposition 2 is loose: covered clusters contribute exactly zero to residual error, not a small positive
 365 quantity. The result strengthens, rather than refutes, the framework. And Pattern A applies precisely
 366 to structural/verifiable classes (format, syntax, schema, invalid proofs), which sit at the head of
 367 the heavy-tailed cluster frequency distribution: the strongest mechanism lands exactly where the
 368 catalogue is densest.

369 **Counter-evidence relocates, not dissolves.** Five prominent steep-decay papers [Dziri et al., 2023,
 370 Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024] are routinely
 371 cited as exponential-compounding evidence. Every one decays over a variable distinct from raw
 372 token length: compositional graph size [Dziri et al., 2023], number of supporting facts [Kuratov
 373 et al., 2024], log-time horizon [Kwa et al., 2025], capacity threshold [Wan et al., 2026], evidence
 374 scope [Karpinska et al., 2024]. The apparent rapid decay is, in each case, in a quantity our framework

already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length: a relocation, not a dissolution. These regimes remain hard; the framework’s value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window expansion that does not help. Appendix G walks through each paper.

Limitations. Postulate 1 is empirical, not derived; a domain with genuinely Heaps-power-law mode discovery would invalidate the doubly-logarithmic special case while preserving the qualitative polylog conclusion. The coverage form of §3.3 is an empirical best-fit; readers should not interpret the numerical match against anchors as theoretical confirmation. Cluster-orthogonality is approximate; Le [2026]’s caveat on prompt-channel interference tightens the bound in shared-channel settings. Domain narrowness: Postulate 1’s empirical anchor rests on three taxonomies (ErrorAtlas, HumanEval, MWPEs), all published 2025–2026 and covering general/code/math; the framework is untested on agentic workflows, long-running scientific reasoning, and multi-turn tool use over millions of tokens, precisely the regimes where $|C|$ may grow faster than logarithmically. The $\sigma \approx 1.85$ estimate is a single-point calibration, not a discovery-curve fit; a subsample-vs-distinct-modes plot has not been published for any LLM error taxonomy at this writing. β is latent and we report no empirical range. Taxonomy granularity (the L2–L3 gap) is unmeasured. Patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously; libraries calibrated against one patch under-cover the next. Finally, the framework *relocates* the difficulty of long-context reliability rather than resolving it: where k_{hard} grows with task length (adversarial compositional structure, multi-hop chains, long agent horizons), reliability remains hard. The contribution is to name the on-axis intervention, not to make those regimes easy. A falsifiability test the framework passes: DebugBench [Tian et al., 2024] explicitly catalogues which classes execution-feedback can and cannot repair, and the empirical split (works for syntax/reference; “unhelpful for logic errors”) is exactly what the framework’s selectivity claim predicts.

7 Conclusion

LLM reliability is often framed as an asymptotic scaling problem. Universal reliability is not a finite-library problem (Proposition 1): an unbounded domain that keeps producing intervention-distinguishable failures cannot be covered by any finite dictionary. The contribution of this paper is to show that, once a bounded deployment patch is fixed, reliability becomes a local engineering problem. Prior work in this line [Anonymous, 2025] argued that the standard asymptotic framing rests on a false uniformity assumption: errors concentrate at $\approx 5\text{--}10\%$ of tokens. This paper takes the next step: within that sparse set, errors are not only concentrated but also *repetitive*, clustering into a finite catalogue whose size grows logarithmically with observed failures under the empirical postulate of §3.2, or as a small power under the conservative Heaps alternative (Appendix E). The conditional consequence (Proposition 2) is that the intervention budget required to bound per-hard-token residual error scales polylogarithmically in sequence length within a fixed domain patch, and becomes a domain-constant once the patch ceiling $|C_D|$ is reached. Sequence-level reliability targets are strictly tighter. Available evidence is consistent with libraries on the order of tens of interventions covering the head of the per-hard-token failure distribution in many fixed domains. The exact number is patch-indexed and should be estimated from local discovery and rank-coverage curves, not assumed from cross-domain priors. This reframes the question from “can we bound the growth of n -token error?” to “have we catalogued enough failure modes *inside the deployment patch*?” The latter is finite and addressable once a bounded deployment patch has been fixed. Direct empirical measurement of the mode-rate constant σ on new domains (agentic, scientific, code-in-production) is the open empirical question this paper invites; future work flagged by Anonymous [2025, §6] on attention-mechanism bounds and stratified-manifold geometry would replace the empirical postulate with a derived result.

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600 A Proof of Proposition 1: No Universal Finite Intervention Dictionary

601 Fix a residual-error tolerance ε . Say that an intervention *covers* a failure mode if it reduces the residual
602 error of that mode below ε . Coverage is mode-level: an intervention that covers a mode covers every
603 failure event in that mode. The two conditions “ D is intervention-unbounded” and “ $|C_D^\varepsilon| = \infty$ ” are
604 equivalent under mode-level coverage; an unbounded witnessing sequence is constructed by taking
605 one representative per mode, and an infinite catalogue forces the existence of such a sequence.

606 Let D be a domain. Call D *intervention-unbounded* if it contains an infinite sequence of failures
607 $f_1, f_2, f_3, \dots$ such that each new f_j is not covered by any finite intervention dictionary that covers
608 all earlier failures $\{f_1, \dots, f_{j-1}\}$.

609 **Step 1: assume the opposite.** Suppose, for contradiction, that C_D^ε is finite. Then there are only
610 finitely many intervention-distinguishable failure modes in D . Write them as $C_D^\varepsilon = \{c_1, \dots, c_M\}$
611 for some finite M .

612 **Step 2: what finiteness means.** If there are only M intervention-distinguishable modes, then after
 613 all M modes have appeared in the sequence, every later failure must belong to one of the already-seen
 614 modes.

615 **Step 3: same mode means same intervention class.** Modes are defined at intervention resolution
 616 ε . If a later failure belongs to the same mode as an earlier failure, then the intervention dictionary that
 617 covers the earlier representative of that mode also covers the later failure below residual tolerance ε .

618 **Step 4: contradiction.** Intervention-unboundedness says exactly the opposite: each new f_j is not
 619 covered by any finite dictionary that covers $\{f_1, \dots, f_{j-1}\}$. Therefore f_j cannot belong to any earlier
 620 intervention mode, so each f_j introduces a new intervention-distinguishable mode. The sequence
 621 $f_1, f_2, f_3, \dots$ induces infinitely many such modes, contradicting the assumption that C_D^ε is finite.
 622 Hence $|C_D^\varepsilon| = \infty$.

623 **Step 5: no finite dictionary covers the domain.** Suppose, again for contradiction, that some finite
 624 intervention dictionary $\mathcal{I}$ covers all of D . Then $\mathcal{I}$ covers every finite prefix $\{f_1, \dots, f_{j-1}\}$ for every
 625 j . By intervention-unboundedness, any dictionary covering that prefix fails to cover f_j . Therefore $\mathcal{I}$
 626 does not cover f_j , contradicting the assumption that $\mathcal{I}$ covers all of D . $\square$

627 B Derivation of Proposition 2: Patch-Local Intervention Budget

628 **Conditions used.** Proposition 2 depends on four assumptions:

629 1. **Coverage model.** The cumulative hard-error mass covered by the top m modes is approximated
 630 by

$$F(m; |C|) = \min\left(1, \frac{\ln m}{\ln |C|}\right)$$

631 on the declared domain $m \geq 2, |C| \geq 2$.

632 2. **Non-trivial, attainable target.** The residual target satisfies $\varepsilon \in (0, e_{\text{hard}})$. If $\varepsilon \geq e_{\text{hard}}$ no
 633 intervention is needed; if $\varepsilon \leq 0$ the target is unattainable unless all residual hard-token error is
 634 eliminated.

635 3. **Patch-local catalogue.** The result applies only after a deployment patch D has been fixed and
 636 its reachable catalogue is modelled as finite or effectively capped.

637 4. **Sequence-length claim.** The doubly-logarithmic rate further requires Assumption 2 and $k(n) =$
 638 $\Theta(\log n)$. Without these, Proposition 2 still gives a catalogue-size budget but not the same
 639 sequence-length scaling.

640 **Step 1: define the target.** Let e_{hard} be the baseline hard-token error rate and $\varepsilon \in (0, e_{\text{hard}})$ the target
 641 after intervention.

642 **Step 2: define the effective catalogue.** For per-sequence scaling, set $C_{\text{eff}} = C_{\text{active}, D}(n)$: how
 643 many failure modes can a single sequence of length n activate? For full deployment-library budgeting,
 644 set $C_{\text{eff}} = C_D$: how large must the library be to cover the recurring failure modes reachable inside
 645 D ?

646 **Step 3: cumulative coverage.** Let the ranked local failure modes have hard-error masses $p_1 \geq$
 647 $p_2 \geq \dots \geq p_{|C_{\text{eff}}|}$ with $\sum_i p_i = 1$. A library covering the top m modes removes cumulative
 648 hard-error mass $F(m; |C_{\text{eff}}|) = \sum_{i=1}^m p_i$, approximated by the log-coverage form of §3.3.

649 **Step 4: residual after intervention.** The uncovered mass fraction is $1 - F(m; |C_{\text{eff}}|)$, so the
 650 residual per-hard-token error rate is $e_{\text{res}}(m) = (1 - F(m; |C_{\text{eff}}|)) e_{\text{hard}}$.

651 **Step 5: impose the target.** Requiring $e_{\text{res}}(m) \leq \varepsilon$ and dividing by $e_{\text{hard}} > 0$ gives

$$1 - F(m; |C_{\text{eff}}|) \leq \frac{\varepsilon}{e_{\text{hard}}}, \quad \text{i.e.,} \quad F(m; |C_{\text{eff}}|) \geq 1 - \frac{\varepsilon}{e_{\text{hard}}}.$$

Step 6: substitute the log-coverage form. Since $\varepsilon < e_{\text{hard}}$, the required coverage $1 - \varepsilon/e_{\text{hard}} \in (0, 1)$, so the cap in F is inactive before saturation. Using $F = \ln m / \ln |C_{\text{eff}}|$:

$$\frac{\ln m}{\ln |C_{\text{eff}}|} \geq 1 - \frac{\varepsilon}{e_{\text{hard}}}.$$

Step 7: solve for m . Multiplying by $\ln |C_{\text{eff}}| > 0$ and exponentiating,

$$m \geq \exp \left[\left(1 - \frac{\varepsilon}{e_{\text{hard}}} \right) \ln |C_{\text{eff}}| \right] = |C_{\text{eff}}|^{1 - \varepsilon/e_{\text{hard}}}.$$

Taking the ceiling (since m is integer-valued) yields $m \geq \lceil |C_{\text{eff}}|^{1 - \varepsilon/e_{\text{hard}}} \rceil$, the bound stated in Eq. (4).

Step 8: sequence-length rate. For per-sequence scaling, set $C_{\text{eff}} = C_{\text{active},D}(n)$. Assumption 2 bounds $|C_{\text{active},D}(n)| \leq \min(A'_D + \sigma'_D \ln h(n), |C_D|)$. In the pre-cap regime, i.e., while $A'_D + \sigma'_D \ln h(n) < |C_D|$, the ceiling has not been reached and $|C_{\text{active},D}(n)| = O(\ln h(n))$. With $h(n) = \beta k(n)$ and $k(n) = \Theta(\log n)$, we get $h(n) = \Theta(\log n)$ and $\ln h(n) = \Theta(\log \log n)$, hence

$$|C_{\text{active},D}(n)| = O(\log \log n),$$

and substituting into the budget,

$$m = O((\log \log n)^{1 - \varepsilon/e_{\text{hard}}}).$$

This is the doubly-logarithmic pre-cap special case.

Step 9: cap regime. Once active-mode discovery saturates the patch catalogue, $|C_{\text{active},D}(n)| = |C_D|$, so $m \geq \lceil |C_D|^{1 - \varepsilon/e_{\text{hard}}} \rceil$, which no longer depends on n . The intervention budget is then domain-constant. $\square$

C Sequence-Level Reliability Derivation

Proposition 2 gives a per-hard-token residual target. Production systems often care about a stricter target: the probability that the entire sequence is correct.

Starting from the composed reliability of Eq. (1), and writing the post-intervention hard-token rate as e_{res} ,

$$P(\text{correct}) = (1 - e_{\text{res}})^{\beta k} (1 - e_{\text{easy}})^{(1 - \beta)k} (1 - e_{\text{non}})^{n - k}.$$

Define the non-hard-token survival factor

$$S_{\text{base}} = (1 - e_{\text{easy}})^{(1 - \beta)k} (1 - e_{\text{non}})^{n - k}, \quad (5)$$

so that $P(\text{correct}) = (1 - e_{\text{res}})^{\beta k} S_{\text{base}}$. The sequence-level target $P(\text{correct}) \geq 1 - \varepsilon_{\text{seq}}$ becomes

$$(1 - e_{\text{res}})^{\beta k} S_{\text{base}} \geq 1 - \varepsilon_{\text{seq}}, \quad \text{i.e.,} \quad (1 - e_{\text{res}})^{\beta k} \geq \frac{1 - \varepsilon_{\text{seq}}}{S_{\text{base}}}.$$

Three regimes follow.

Regime (i): non-hard-token errors already violate the target. If $S_{\text{base}} < 1 - \varepsilon_{\text{seq}}$, then even setting $e_{\text{res}} = 0$ cannot meet the target, because the maximum possible survival after eliminating all hard-token failures is only S_{base} . Hard-token interventions alone cannot meet the sequence-level SLA. This is the route by which the exponential-in- n concern re-enters and should be diagnosed before any catalogue-budgeting exercise.

Regime (ii): hard-token residual error determines feasibility. If $S_{\text{base}} \geq 1 - \varepsilon_{\text{seq}}$, the target may be feasible. Taking the $(1/(\beta k))$ -th power of the rearranged inequality and isolating e_{res} ,

$$e_{\text{res}} \leq 1 - \left(\frac{1 - \varepsilon_{\text{seq}}}{S_{\text{base}}} \right)^{1/(\beta k)} =: \tau_{\text{seq}}.$$

Applying Proposition 2 with ε replaced by τ_{seq} ,

$$m \geq \lceil |C_{\text{eff}}|^{1 - \tau_{\text{seq}}/e_{\text{hard}}} \rceil.$$

As the sequence-level target becomes stricter (ε_{seq} shrinks), $\tau_{\text{seq}} \rightarrow 0$ and the exponent $1 - \tau_{\text{seq}}/e_{\text{hard}} \rightarrow 1$, so $m \rightarrow |C_{\text{eff}}|$. Strict one-shot sequence-level reliability pushes the system toward full-catalogue coverage.

684 **Regime (iii): baseline hard-token error is already acceptable.** If $\tau_{\text{seq}} \geq e_{\text{hard}}$, the baseline hard-
685 token rate already satisfies the sequence target (no intervention is required because $e_{\text{res}}(0) = e_{\text{hard}}$
686 from Eq. (3)).

687 **Conclusion.** Per-hard-token reliability is easier than sequence-level reliability. A library that gives
688 a large reduction in residual hard-token error may still be insufficient for one-shot sequence-level
689 guarantees when many hard decisions occur in a single output. This is why the main paper treats the
690 “ ≈ 50 patterns” rule of thumb as a per-hard-token planning prior, not a one-shot sequence-level SLA.

691 D Why these are propositions rather than unconditional theorems

692 Proposition 1 is a definitional impossibility result: once intervention-unboundedness is assumed, an
693 infinite intervention-resolution catalogue follows. Its role is not to prove that every unbounded domain
694 necessarily has infinite failure modes, but to show that open-ended domains cannot be assumed to
695 admit finite dictionaries.

696 Proposition 2 is conditional engineering math. It does not prove that LLM failures universally obey
697 logarithmic mode discovery. It proves that *if* a bounded patch has a finite or effectively capped
698 reachable catalogue, *and if* cumulative intervention coverage follows the stated head-heavy form, *then*
699 the required per-hard-token intervention budget grows slowly and becomes constant after catalogue
700 saturation.

701 The empirical burden therefore lies not in the algebra but in measuring, for each deployment patch, the
702 local discovery curve $C_{\text{seen},D}(T)$, the per-sequence activation $C_{\text{active},D}(n)$, the cumulative coverage
703 $F(m; |C_D|)$, the hard-token fraction β_D , and the baseline hard-token rate e_{hard} . This is why the paper
704 frames the results as a reliability-engineering scaffold rather than as universal theorems about LLM
705 behaviour.

706 E Heaps Power-Law Variant and Cluster-Count Sensitivity

707 A reader who prefers to derive cluster-count growth from standard Heaps’ law rather than from
708 Postulate 1 obtains a qualitatively similar result. Let $|C|(k_{\text{hard}}) \approx K \cdot k_{\text{hard}}^b$ with $b \in (0, 1)$. Canonical
709 fits give $b \approx 0.5$ for natural-language vocabularies; failure-mode taxonomies plateau much more
710 sharply (ErrorAtlas stabilises at $|C| = 17$ across 10^4+ failures), implying a smaller effective $b \approx 0.2$
711 in our setting. Composing with $k = \Theta(\log n)$, we get $|C| = O((\log n)^b)$, and Proposition 2 becomes

$$m = O\left((\log n)^{b \cdot (1 - \varepsilon / e_{\text{hard}})}\right).$$

712 This is still polylogarithmic in n for any $b \in (0, 1)$ and any $\varepsilon < e_{\text{hard}}$. The paper’s qualitative claim
713 survives either choice of cluster-count law.

714 **Symbolic-form sensitivity across candidate laws.** The polylog conclusion depends on which
715 cluster-count law one accepts; available evidence is consistent with multiple candidates because no
716 subsample-discovery curve has been published for any LLM failure-mode taxonomy at this writing.
717 We therefore report symbolic rates rather than fitted constants. With $h(n) = \beta k(n)$:

- 718 • **Logarithmic:** $|C_{\text{active},D}(n)| = O(\log h(n))$. If $k(n) = \Theta(\log n)$, then $m =$
719 $O((\log \log n)^{1 - \varepsilon / e_{\text{hard}}})$.
- 720 • **Heaps:** $|C_{\text{active},D}(n)| = O(h(n)^b)$ with $b \in (0, 1)$. If $k(n) = \Theta(\log n)$, then $m =$
721 $O((\log n)^{b(1 - \varepsilon / e_{\text{hard}})})$.
- 722 • **Saturating:** $|C_{\text{active},D}(n)| \leq |C_D|$. Then $m = O(|C_D|^{1 - \varepsilon / e_{\text{hard}}})$, constant in n once the
723 patch ceiling is reached.

724 The qualitative conclusion is robust **under the $k(n) = \Theta(\log n)$ regime:** m grows more slowly than
725 any positive power of n under every candidate, and the directional claim (“a small library covers the
726 head of the failure distribution in the per-hard-token regime”) survives. Only the exponent shifts:
727 doubly-logarithmic under logarithmic discovery, $(\log n)^b$ with small b under Heaps, constant in the

cap regime. The doubly-logarithmic rate is the optimistic special case. If $k(n)$ grows as a positive power of n , the Heaps variant inherits that power and the polylog-in- n language fails along that axis; the framework’s intervention prescription still applies, but its asymptotic-rate framing does not.

Numerical constants require a measured discovery curve $C_{\text{seen},D}(T)$ or $C_{\text{active},D}(n)$. Existing taxonomies provide endpoint category counts at a single corpus scale (ErrorAtlas at $|C| = 17$ for $\approx 10^4$ failures), not discovery curves. The subsample-discovery measurement remains the explicit empirical test that would either tighten the postulate or fall back to the Heaps variant. Until that measurement exists, the framework’s headline rate should be read as *polylogarithmic in the pre-cap regime, domain-constant in the cap regime*, with the specific exponent flagged as a falsifiability test rather than a fitted prediction.

F Full Failure-Mode Taxonomy and the Capability-Elimination Harvest

The intervention literature provides at least one targeted countermeasure for each named cluster of §4. Two organising granularities exist: at the capability level (six axes) the cluster-orthogonality property is most clearly visible; at the error-class level (twelve named clusters) the evidence aligns with the taxonomies of §4 but is finer-grained than the underlying capability mechanisms.

Six capability axes and three structural patterns. A dedicated harvest yields **28 quantitatively-anchored citations** across **six independent capability axes**: Arithmetic (Python/symbolic execution), Code Execution (REPL/sandbox feedback), Format/Structure (constrained decoding, FSMs, grammar engines), Perception/Grounding (visual grounding for GUI, charts, tables), Knowledge/RAG (dense retrieval and citation grounding), Verification (proof checkers, learned verifiers, classifier rerouting, process supervision). Each axis is independently confirmed by between three and nine citations. We stratify the 28 by kind of evidence into three patterns:

Pattern A: hard guarantees (by-construction). Seven citations achieve residual error rate equal to zero *by construction*, restricted strictly to structural/verifiable classes: constrained decoders set $P(\text{invalid token}) = 0$ at every step [Suresh et al., 2025, Zhang et al., 2023, Dong et al., 2025, Li et al., 2026, OpenAI, 2024]; static syntax checks reject programs with a `SyntaxError` before execution [Wen et al., 2024]; proof kernels reject any output failing type-checking [Ren et al., 2025]. The class of grammar-violating outputs is mathematically empty under these mechanisms.

Pattern B: strong empirical reductions with class-shift signature. Roughly fourteen citations report empirical reductions of 80–100% in a named error class, with the post-intervention failure log dominated by structurally different residual classes. Program-of-Thoughts on GSM8K [Chen et al., 2023]: calculation errors drop from 30% of failures to 0%, residuals are 62% reasoning + 36% misunderstanding. OpenMedCalc [Goodell et al., 2025]: “only interpretation errors were identified.” Acurai [Wood and Forbes, 2024]: 100% hallucination elimination on RAGTruth (95% CI 91–100%), strong empirical rather than by-construction.

Pattern C: moderate reductions (60–80%) with residuals shifting outside the target class. Legal RAG [Dantart, 2026]: fabricated citations $> 30\% \rightarrow < 0.2\%$. GPT-5 SimpleQA with web access [OpenAI, 2025]: 47% $\rightarrow$ 9.6% (inter-condition, not within-condition). CRITIC [Gou et al., 2024a]: toxic generation -79.2% .

The 28 citations, organised by axis and pattern.

Axis	Citation & targeted class	Pre → Post	Pattern
Verification	Ren et al. [2025]: Invalid Lean proof	any → 0% by constr.	A
Format	OpenAI [2024]: JSON schema violation	>60% → 0% by constr.	A
Format	Zhang et al. [2023]: Tool-call syntax	21–100% → 0%	A
Format	Suresh et al. [2025]: JSON parse failure	13–82% → 0%	A
Format	Dong et al. [2025]: Multi-format errors	20–38% → 0%	A
Format	Li et al. [2026]: Malformed tool calls	33–78% → 0%	A
Arithmetic	Chen et al. [2023]: Calculation on GSM8K	30% → 0% of failures	B
Arithmetic	Goodell et al. [2025]: Clinical arithmetic	“only interpretation errors”	B
Knowledge/RAG	Wada et al. [2025]: RAG hallucinations	8% → 0% ($p = 0.012$)	B
Knowledge/RAG	Wood and Forbes [2024]: Context-conflict hallucinations	100% → 0% on subset	B
Knowledge/RAG	Dantart [2026]: Fabricated legal citations	>30% → <0.2%	C
Code Exec	Wen et al. [2024]: <code>SyntaxError</code> (HumanEval)	5.76% → 0.01%	A
Code Exec	Li et al. [2022]: False-positive submissions	62% → 4%	B
Code Exec	Shi et al. [2024]: 5 code-bug classes	100% repair (5/6 classes)	B
Knowledge/RAG	Gao et al. [2023b]: Citation grounding (ELI5)	≈50% w/o complete support	C
Knowledge/RAG	OpenAI [2025]: Factual errors	47% → 9.6% w/web	C
Knowledge/RAG	Zakka et al. [2024]: Clinical citation errors	44% → 9%	C
Arithmetic	Wang et al. [2025a]: Arithmetic in medical reasoning	426 → 74 errors	B
Code Exec	Wen et al. [2024]: <code>NameError</code> repair	22.7% → 2.3%	C
Perception	Gou et al. [2025]: GUI grounding errors	16.2% → 73.3% acc.	B
Perception	Xie et al. [2025]: GUI task failures	5% → 27% SR ($5.4\times$)	B
Perception	Cheng et al. [2024]: Element mislocation	5.2% → 53.4% ($10.3\times$)	B
Perception	Liu et al. [2023]: Chart-reading errors	38.2% → 67.6% (+29.4 pp)	B
Verification	Gou et al. [2024a]: Toxic generation	−79.2%	C
Verification	Wang et al. [2024b]: Reasoning step errors	28.6% → 43.5% w/rerank	B
Knowledge/RAG	Asai et al. [2024]: Unsupported facts	55.5% → 22% error	B
Perception	Lu et al. [2024]: Icon/element errors	70.5% → 93.8% (79% red.)	B
Knowledge/RAG	Mallen et al. [2023]: Long-tail entity errors	≈80% → ≈50% failure	B

Capabilities are coarser than error classes. Several entries in the harvest reveal “two-for-one” reductions where one capability addresses multiple named clusters: a single Python interpreter removes the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; constrained decoding eliminates by construction both format violations and the structural component of “missing required element”; code execution feedback strongly reduces `SyntaxError`, `NameError`, and most `TypeError` together; RAG strongly reduces factual hallucinations, fabricated citations, and outdated information jointly. The practical capability library required is therefore smaller than the count of named error categories.

The twelve named clusters.

Cluster	Axis	Intervention	Before → After	Patt.
A. Arithmetic	Arithmetic	PAL Python [Gao et al., 2023a]	20.1% → 61.5%	B
B. Unit conversion	Arithmetic	Folded into A (Python)	n/a	n/a
C. Counting	Arithmetic	Explicit counter (gap)	n/a	gap
D. Format/schema	Format	DINGO/SLOT [Suresh et al., 2025, Wang et al., 2025b]	up to +68 pp; 99.5% acc.	A
E. Code logic/type	Code Exec	Reflexion/AgentCoder [Shinn et al., 2023, Huang et al., 2023]	80% → 96.3% pass@1	A/B
F. Multi-hop drift	Knowledge	Entity-grounded rewriter	≈5–10 pp	C
G. Reasoning step	Verification	Math-Shepherd [Wang et al., 2024b]	28.6% → 43.5%	B
H. Spec misinterpret.	Verification	Clarification [Niwa and Iso, 2024]	ROUGE-L +15	C
I.a Struct. missing	Format	Subsumed by D (constr. decode)	n/a	A
I.b Semantic missing	Verification	Subsumed by G/H	n/a	C
J. Hallucination	Knowledge	RAG [Wood and Forbes, 2024]	non-zero → 0%	B/C
K. Refusal	Verification	POROver [Karaman et al., 2024]	57.6% → 82.1%	B
L. Tool/API	Format+Verif.	SAGE-Agent [Suri et al., 2025]	36.5% → 65.2%	B

Eight of twelve categories have a strong citation with double-digit percentage-point improvement; three (B, C, I) lack a clean single-cluster ablation; one (F) has consistent medium-strength evidence. Category I (missing required elements) splits mechanistically into I.a (structural, absorbed by D’s constrained decoders) and I.b (semantic, absorbed by G/H). Categories B (units) and C (counting) remain technical gaps: B is naturally folded into A; C is the smallest residual gap.

Additivity and its limits. Patel et al. [2026] demonstrate that stacking interventions targeting orthogonal failure modes can produce large compound reliability gains: their parallel-consensus framework yields a $14,700\times$ improvement over single-pass baseline, evidence for compound gains from decomposition plus consensus aggregation under their specific parallel-voting regime rather than a direct demonstration that heterogeneous cluster-targeted interventions compose additively without voting. Shang et al. [2024] similarly show supra-additive gains when modules target distinct failure modes. Le [2026] reports that schema-level and prompt-level instructions interact *non-additively* when sharing a prompt channel: interventions on orthogonal processing layers (decoding constraint vs. retrieval vs. training-signal vs. inference-time tool call) compose near-additively, while those sharing a channel may interfere.

The irreducible-semantic residual. Of the 17 ErrorAtlas categories, 13 are addressed under Patterns A, B, or C by one of the six capability axes; four are not: residuals of inappropriate refusal beyond preference optimisation, specification misinterpretation not closed by clarification, problem-decomposition reasoning bottlenecks, and a “user wanted something different” semantic remainder. These are classes where the failure is in choosing what to do, not executing it. Proposition 2’s prediction of $O(\log)$ rather than $O(0)$ residual reliability reflects exactly this irreducible core. The framework does not claim 100% coverage of all failures; it claims polylog-bounded *capability-eliminable* failures, with the named semantic residual as the floor.

803 G Counter-Evidence Re-Audits

Five prominent papers are routinely cited as evidence that LLM reliability decays steeply with length [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024]. A careful re-reading shows that *every one* of these papers decays over a variable distinct from raw token length n . Identifying the decay axis is not the same as dissolving the practical concern: where compositional graph size, fact count, or evidence scope grow with problem length, the framework predicts steep failure curves. The contribution is to identify which interventions help (capability provisioning along the actual decay axis) and which do not.

Dziri et al. [2023], Faith and Fate. GPT-4 multi-digit multiplication accuracy drops from 59% (3-digit) to 4% (4-digit) zero-shot, with the authors theorising “probability of incorrect predictions

converges exponentially to ≈ 1 for abstract compositional tasks.” The decay variable is *compositional graph size* N , not raw token length: the 3×3 graph has on the order of d^2 partial products plus carries, and multi-digit multiplication is engineered so that $k_{\text{hard}} \approx N$, with every node in the computation graph a hard decision. For natural-language tasks where $k_{\text{hard}} \ll n$, Dziri’s regime is the *boundary case* in which our framework reduces to their result. A clean $(1 - \varepsilon)^N$ exponential cannot simultaneously reproduce 59% at 3-digit and 4% at 4-digit for any single per-node ε : the observed drop is locally steeper than per-node iid exponential, consistent with a finite catalogue of failure modes exhausting as N grows. *Where the framework concedes*: in adversarial compositional tasks engineered so $k_{\text{hard}} \approx N$, the framework reduces to Dziri’s regime and does not relieve it; the relocation is informational, not magical.

Kuratov et al. [2024]. The abstract reports “performance declines sharply with increased reasoning complexity” and models “effectively utilise only 10–20% of the context.” Read in detail, BABILong varies two axes: context length n (0K to 10M tokens) and number of supporting facts k (QA1 = 1 fact to QA3 = 3 facts). The sharp decline is in k , not n . For QA1 (single-fact), most models “perform well up to 4,000 tokens”: a plateau, not exponential decay. The famous RAG-flat-across-length result (60% on single-fact QA *independent of context length*) is the cleanest possible demonstration that when relevant evidence is in window, length does not matter. Recurrent Memory Transformers maintaining performance to 50M tokens further confirms effective k is determined by architecture, not raw n . The framework’s concession is narrow but real: for multi-hop tasks whose required fact count grows with task complexity, BABILong’s sharp k -axis decay is exactly what the framework predicts happens, not a counter-example.

Kwa et al. [2025] (METR). Per-model success fits a logistic in $\log(\text{human-task-duration})$: $S(t) = \sigma(\beta(\log t - \log h))$. Logistic-in-log-length is *mathematically sublinear* in length itself. The 80%-horizon being 4–6 $\times$ shorter than the 50%-horizon is a steep within-model cliff but compatible by construction with the polylog result: a cliff at a specific capacity threshold is a manifold-transition signature. The famous exponential (capability doubling every seven months) is an *inter-model* claim about how the horizon h moves across generations, orthogonal to within-model decay shape. An honest qualifier: the within-model logistic-in-log cliff is steep on a practical scale. Calling it “sublinear in length” is technically correct but engineering-useful only for systems designed to operate well below the 50% horizon.

Wan et al. [2026], Fano-style upper bound. This paper theorises super-linear information-demand growth and identifies an “accuracy cliff” at capacity overflow: “when the task’s information demand surpasses the model’s output capacity, performance does not degrade gracefully but instead collapses sharply.” The behavioural prediction is a *threshold*, not smooth $(1 - \varepsilon)^n$. A cliff at a specific capacity threshold is exactly the manifold-transition behaviour our framework predicts at the boundary between covered and uncovered modes; Wan et al.’s mechanism (capacity overflow) is one specific cause, complementary to ours. *Where the framework concedes*: Wan documents a mechanism by which $|C|$ effectively exceeds the model’s coverage in a single forward pass; Pattern A interventions (constrained decoding, formal verification) are the kind of response the framework expects but does not automatically supply.

Karpinska et al. [2024]. GPT-4o achieves 55.8% pair accuracy across 1,001 minimally-different true/false claim pairs about long fictional books (mean length 127K tokens). The paper reports performance by *evidence scope*, not by context length: 59.8% on sentence-level retrieval, 47.6% on passage-level, 41.6% on global reasoning. No per-context-length curve is reported. The decay axis is the number of evidence pieces that must be integrated, a k -axis quantity rather than an n -axis one. NoCha is therefore not counter-evidence to a sublinear-in- n claim. Evidence-scope decay is exactly the k -axis observation the framework predicts cannot be addressed by scaling raw context length; it requires retrieval or process supervision along the actual decay axis. The relocation is informational, not magical.

Unifying observation. Every counter-paper decays over a variable that is not n : compositional graph size, fact count, log-time horizon, capacity threshold, or evidence scope. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length. This is a relocation, not a dissolution. Where k_{hard} grows with task length (adversarial compositional structure, multi-hop fact chains, long horizons that force more

867 decisions), reliability remains hard; the framework's value is directing intervention toward capability
868 provisioning along the actual decay axis rather than toward context-window or compute-budget
869 expansion that does not help.

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